Loxodromic Sequences of Tangent Spheres

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The first three sections add one more to the considerable number of published proofs of the Descartes circle theorem [10] and of Gosset's extension to Euclidean n-space. The last five sections apply these results to the study of an infinite sequence of spheres in n-space having the property that every $n+2$ consecutive members are mutually tangent. It will be proved that, for every $n>1$, just one such sequence exists with radii in geometric progression. The ratio of the G.P. steadily approaches 1 as n tends to infinity. This conclusion is not unexpected, as Hilbert space admits a countable infinity of congruent spheres, mutually tangent.

I am grateful to J. B. WILKER for drawing the figures.

1. The Descartes Circle Theorem

In November 1643, DESCARTES wrote a letter to his favorite disciple, Princess ELISABETH OF BOHEMIA. This letter has been preserved [5, pp. 45–50]. It includes a sketch of four circles in mutual external contact. He denotes the radii by d, e, f, x and seeks an equation connecting them. Observing that the four centres have mutual distances

$$d+x, \quad e+x, \quad f+x, \quad e+f, \quad f+d, \quad d+e,$$

he obtains by 'brute force' the equation

$$\begin{aligned} ddeeff + ddeexx + ddffxx + eeffxx &= 2deffxx \\ &+ 2deeffx + 2deefxx + 2ddeffx + 2ddefxx + 2ddeefx, \end{aligned}$$

which PEDOE [10, p. 634] proposes to call 'the Descartes circle theorem' [cf. 2, pp. 13–16]. BLUMENTHAL [1a, pp. 153–154] gives a neat proof for the determinantal relation that connects the mutual distances of four points. (For the extension to n dimensions, see [15, p. 125].) As AEPPLI remarks [1, p. 10], this yields, for the four centres,

$$\begin{vmatrix} 0 & (d+e)^2 & (d+f)^2 & (d+x)^2 & 1 \\ (e+d)^2 & 0 & (e+f)^2 & (e+x)^2 & 1 \\ (f+d)^2 & (f+e)^2 & 0 & (f+x)^2 & 1 \\ (x+d)^2 & (x+e)^2 & (x+f)^2 & 0 & 1 \\ 1 & 1 & & 1 & 0 \end{vmatrix} = 0,$$

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$$\begin{vmatrix} -d^2 & de & df & dx & 1 \\ ed & -e^2 & ef & ex & 1 \\ fd & fe & -f^2 & fx & 1 \\ xd & xe & xf & -x^2 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0, \qquad \begin{vmatrix} -1 & 1 & 1 & 1 & \varepsilon_1 \\ 1 & -1 & 1 & 1 & \varepsilon_2 \\ 1 & 1 & -1 & 1 & \varepsilon_3 \\ 1 & 1 & 1 & -1 & \varepsilon_4 \\ \varepsilon_1 & \varepsilon_2 & \varepsilon_3 & \varepsilon_4 & 0 \end{vmatrix} = 0,$$

where $\varepsilon_1 = 1/d$, $\varepsilon_2 = 1/e$, $\varepsilon_3 = 1/f$, $\varepsilon_4 = 1/x$. The case when one circle encloses the other three can be covered by allowing one of d, e, f, x (or one of the curvatures ε_ν) to be negative. The limiting case when one or two circles reduce to straight lines can be covered by allowing one or two ε's to be zero.

PEDOE [10, p. 630] relieves us from the task of evaluating the last determinant (or its companion on the left, as Descartes, in effect, did) by remarking that this determinant, being quadratic and symmetric in the ε's, must be equal to

$$a(\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \varepsilon_4^2) + b(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)^2$$

for certain constants a and b; the instance $\varepsilon_1 = \varepsilon_2 = 1$, $\varepsilon_3 = \varepsilon_4 = 0$ (Figure 1) shows that the equation must actually be

$$2(\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \varepsilon_4^2) - (\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)^2 = 0. \tag{1.1}$$

In the form $\sum \varepsilon^2 = \frac{1}{2}(\sum \varepsilon)^2$, this agrees with two memorable lines of SODDY's poem [13]:

The sum of the squares of all four bends
Is half the square of their sum.

In applying (1.1), we must carefully remember that, if two of the four mutually tangent circles lie on the same side of the tangent at their point of contact, the smaller curvature (or 'bend') is negative. For instance [11], we may have

$$\varepsilon_1 = 6, \quad \varepsilon_2 = 3, \quad \varepsilon_3 = 2, \quad \varepsilon_4 = -1.$$

Another interesting proof was given by STEINER [16, pp. 161, 274–275]. He expressed ε_4 in terms of ε_1, ε_2, ε_3, but did not write out the symmetric relation between the four bends.

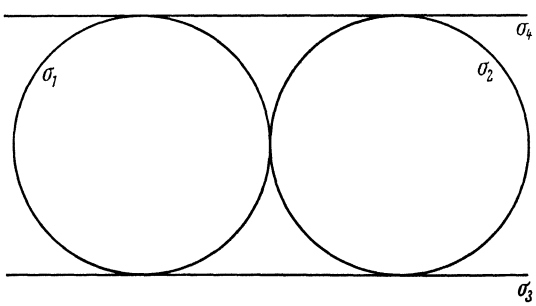

Figure 1

2. A New Proof Using Inversion

Following MORLEY [9], we use two simple lemmas. (See [4, p. 30] for the notion of *power*.)

With respect to a circle σ of radius r, let p be the power of an outside point C. Then the circle with centre C and radius k inverts σ into a circle of radius $k^2 r/p$. (2.1)

Proof [17, pp. 401–402]. Let d be the distance from C to the centre of σ. Observing that the diameter of σ which passes through C is mapped by the inversion onto the diameter of the image circle, we find that the diameter of the inverse circle is

$$\frac{k^2}{d-r} - \frac{k^2}{d+r} = \frac{2k^2 r}{d^2 - r^2} = \frac{2k^2 r}{p}.$$

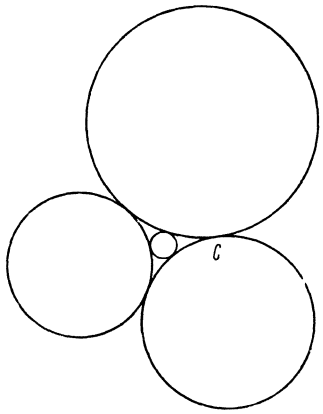

Figure 2

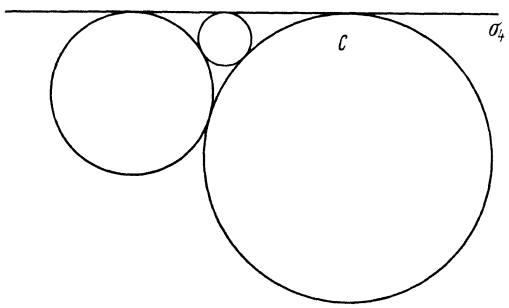

Figure 3

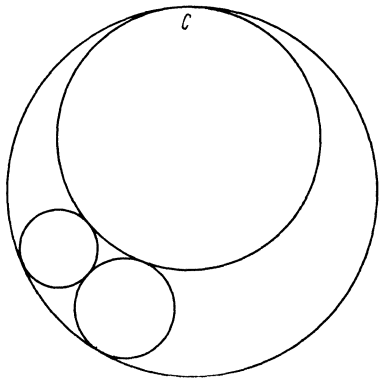

Figure 4

$$\left.\begin{array}{l}\textit{Let } b \textit{ be the distance from a point } C \textit{ to a line } l. \textit{ Then the circle}\\ \textit{with centre } C \textit{ and radius } k \textit{ inverts } l \textit{ into a circle (through } C\textit{) of}\\ \textit{radius } k^2/2b.\end{array}\right\} \tag{2.2}$$

Proof. Obvious.

Proof of the Descartes circle theorem (1.1). Let the four mutually tangent circles be so numbered that $\varepsilon_1 \geqslant \varepsilon_2 \geqslant \varepsilon_3 \geqslant \varepsilon_4$. If $\varepsilon_3 = \varepsilon_4 = 0$, as in Figure 1, we have two parallel lines, so that $\varepsilon_1 = \varepsilon_2$. In every other case we have $\varepsilon_1 \geqslant \varepsilon_2 \geqslant \varepsilon_3 > 0$ and either $\varepsilon_4 > 0$ (Figure 2) or $\varepsilon_4 = 0$ (Figure 3) or $\varepsilon_4 < 0$ (Figure 4). Let C be the point of contact of the third and fourth circles. The circle with centre C and radius k inverts these last two circles into two parallel lines, σ_3 and σ_4, and inverts the first and second circles into two congruent circles, σ_1 and σ_2, which touch these lines and each other, as in Figure 1. (So far, this is the same procedure that PEDOE used [10, pp. 631–633]. He mentions Lemma 2.1 as 'what should be a well-known formula'. The new twist involves a different choice of coordinates.)

For a suitable value of k, the congruent circles σ_1 and σ_2 have radius 1, and we can choose coordinate axes so that σ_1, σ_2, and σ_3, σ_4 are

$$x^2 + y^2 \pm 2x = 0 \quad \text{and} \quad y \pm 1 = 0.$$

The original four circles may be regarded as arising from these by the same inversion, that is, by inversion in a circle of radius k whose center C is, say, (x, y). In the situations of Figures 2, 3, and 4 we have $-1 < y < 1$, $y = 1$, $y > 1$, respectively; thus ε_4 has the same sign as $-y+1$.

By Lemma 2.1 with $r = 1$ and $p = x^2 + y^2 \pm 2x$,

$$\left.\begin{array}{l}\varepsilon_1\\ \varepsilon_2\end{array}\right\} = \frac{x^2 + y^2 \pm 2x}{k^2}.$$

By Lemma 2.2,

$$\left.\begin{array}{l}\varepsilon_3\\ \varepsilon_4\end{array}\right\} = \frac{2(\pm y + 1)}{k^2}.$$

Hence

$$\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 = 2\,\frac{x^2 + y^2 + 2}{k^2}$$

and

$$\begin{aligned}\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^3 + \varepsilon_4^4 &= 2\,\frac{(x^2 + y^2)^2 + 4x^2 + 4(y^2 + 1)}{k^4}\\ &= 2\left(\frac{x^2 + y^2 + 2}{k^2}\right)^2 = \tfrac{1}{2}(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)^2.\end{aligned}$$

3. Gosset's Theorem

The Descartes circle theorem was extended to 3 dimensions by SODDY in the third verse of his poem [13] and to n dimensions by GOSSET in another poem [7]. The latter can be paraphrased as follows:

In Euclidean n-space $(n \geqslant 2)$, *let* $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_{n+2}$ *be the bends of* $n+2$ *mutually tangent spheres. Then*

$$n \sum \varepsilon^2 - (\sum \varepsilon)^2 = 0. \tag{3.1}$$

By 'spheres' we mean, of course, 'hyperspheres' or '$(n-1)$-spheres'. Since the word 'curvature' is ambiguous when $n > 2$, it is very convenient to follow SODDY and GOSSET in defining the *bend* of any sphere to be the reciprocal of its radius, with a minus sign in appropriate cases.

The following new proof is valid for all $n \geqslant 2$, but does not precisely reduce to the proof in § 2 when we set $n=2$. Let $(x_1, \ldots, x_n)$ be the point of contact C of the two largest spheres. Then C is outside each of the remaining n spheres, and the whole set of $n+2$ may be regarded as arising by inversion from n mutually tangent spheres $\sigma_1, \ldots, \sigma_n$, sandwiched between two parallel hyperplanes σ_{n+1} and σ_{n+2}. It follows that the spheres $\sigma_1, \ldots, \sigma_n$ are all congruent, and their centres are the vertices of a regular simplex.

We choose the coordinate system so that σ_ν $(\nu \leqslant n)$, having centre $(0, \ldots, 1, 0, \ldots, 0)$ and radius $1/\sqrt{2}$, is

$$\sum x^2 - 2x_\nu + \tfrac{1}{2} = 0,$$

while σ_{n+1} and σ_{n+2} are the parallel hyperplanes

$$\sum x - 1 \pm \frac{\sqrt{n}}{\sqrt{2}} = 0, \quad \text{or} \quad \frac{\sum x - 1}{\sqrt{n}} \pm \frac{1}{\sqrt{2}} = 0.$$

By Lemma 2.1, in which the circles can be replaced by spheres,

$$\varepsilon_\nu = \frac{p}{k^2 r} = \frac{\sqrt{2}}{k^2} (\sum x^2 - 2x_\nu + \tfrac{1}{2}) \quad (\nu \leqslant n).$$

By Lemma 2.2, since the ε's are all positive when C lies between the two parallel hyperplanes,

$$\varepsilon_{n+1} = \frac{2}{k^2}\left(\frac{\sum x - 1}{\sqrt{n}} + \frac{1}{\sqrt{2}}\right), \quad \varepsilon_{n+2} = \frac{2}{k^2}\left(-\frac{\sum x - 1}{\sqrt{n}} + \frac{1}{\sqrt{2}}\right).$$

Hence

$$\sum \varepsilon = \frac{\sqrt{2}}{k^2}\left(n \sum x^2 - 2 \sum x + \frac{n}{2} + 2\right)$$

and

$$
\begin{aligned}
n\sum\varepsilon^2 &= \frac{2n}{k^4}\{n(\textstyle\sum x^2+\frac{1}{2})^2 - 4(\sum x^2+\frac{1}{2})\sum x + 4\sum x^2\} + \frac{8}{k^4}\left\{(\sum x-1)^2+\frac{n}{2}\right\}\\
&= \frac{2}{k^4}\left\{n^2(\sum x^2)^2 + n^2\sum x^2 + \frac{n^2}{4} - 4n\sum x^2\sum x\right.\\
&\quad \left. - 2n\sum x + 4n\sum x^2 + 4(\sum x)^2 - 8\sum x + 4 + 2n\right\}\\
&= \frac{2}{k^4}\left\{n^2(\sum x^2)^2 + 4(\sum x)^2 + \left(\frac{n}{2}+2\right)^2 - 4n\sum x^2\sum x\right.\\
&\quad \left. + n(n+4)\sum x^2 - 2(n+4)\sum x\right\}\\
&= \frac{2}{k^4}\left(n\sum x^2 - 2\sum x + \frac{n}{2} + 2\right)^2\\
&= (\sum\varepsilon)^2 .
\end{aligned}
$$

Although this proof of (3.1) is less laborious than that of WISE [18] and FOG [6], it still involves some tiresome algebraic manipulation and lacks symmetry (in that two of the $n+2$ spheres play a special role). Both these drawbacks will be avoided if we can invert any $n+2$ mutually tangent spheres into $n+2$ congruent spheres. This is, of course, impossible for $(n-1)$-spheres in n dimensions; so let us work in $n+1$ dimensions.

Any set of $n+2$ mutually tangent $(n-1)$-spheres, in a Euclidean n-space ϱ, may be regarded as the 'equatorial' section of a set of $n+2$ mutually tangent n-spheres in a Euclidean $(n+1)$-space. By inverting any such set of n-spheres in an n-sphere whose centre is one of the points of contact, we obtain two parallel hyperplanes (that is, n-spaces) with n congruent n-spheres sandwiched between them. These n have their centres at the vertices of an $(n-1)$-dimensional regular simplex α_{n-1}. Therefore all sets of $n+2$ mutually tangent n-spheres are equivalent under the group generated by all inversions. In particular, any such set is inversively equivalent to a set of $n+2$ congruent n-spheres with their centres at the vertices of a regular simplex α_{n+1}. In this instance, the $n+2$ n-spheres all touch two other n-spheres (one 'big' and one 'small'), concentric with the simplex. Although concentric spheres do not necessarily invert into concentric spheres, non-intersecting spheres do remain non-intersecting. Hence any $n+2$ mutually tangent n-spheres touch two non-intersecting n-spheres.

Let us now apply this result to the $n+2$ mutually tangent n-spheres whose centres lie in the hyperplane ϱ. The two non-intersecting n-spheres that touch them have their centres on a line l (perpendicular to ϱ) which contains two *limiting points*. These are the common points of the bundle of circles orthogonal to the two n-spheres (appearing as an intersecting pencil of coaxal circles in each plane through l). Either of the

limiting points can be used as the centre C of an n-sphere that will invert the bundle into a bundle of lines, and the two n-spheres into *concentric* n-spheres. In this manner, the $n+2$ mutually tangent n-spheres will be inverted into congruent n-spheres around the vertices of a regular simplex α_{n+1}. The $\binom{n+2}{2}$ points of contact of pairs of these $n+2$ congruent n-spheres are, of course, the midpoints of the $\binom{n+2}{2}$ edges of the regular simplex. Hence these points all lie on an n-sphere ϱ' which, being orthogonal to each of the $n+2$ congruent n-spheres, is precisely the inverse of the original hyperplane ϱ. Hence C lies on ϱ'.

Using an $(n+2)$-dimensional coordinate frame for the sake of symmetry, let

$$(1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$$

be the centres of the $n+2$ congruent n-spheres in the $(n+1)$-space $\sum x=1$, meaning

$$x_1 + \cdots + x_{n+2} = 1 .$$

Then the n-spheres themselves are the sections, by this hyperplane, of the $(n+1)$-spheres

$$\sum x^2 - 2x_\nu + \tfrac{1}{2} = 0 \qquad (\nu = 1, \ldots, n+2)$$

of radius $1/\sqrt{2}$. Since a typical point of contact is $(\frac{1}{2}, \frac{1}{2}, 0, \ldots, 0)$, the orthogonal n-sphere ϱ' is the section by $\sum x=1$ of the $(n+1)$-sphere $\sum x^2=\frac{1}{2}$. Thus the coordinates $(x_1, \ldots, x_{n+2})$ of C satisfy

$$\sum x = 1 \quad \text{and} \quad \sum x^2 = \tfrac{1}{2} .$$

By Lemma 2.1 with $r=1/\sqrt{2}$ and $p=\sum x^2-2x_\nu+\frac{1}{2}$,

$$\varepsilon_\nu = \frac{\sqrt{2}}{k^2}\left(\sum x^2 - 2x_\nu + \tfrac{1}{2}\right) = \frac{\sqrt{2}}{k^2}(1 - 2x_\nu) .$$

Hence

$$\sum \varepsilon = \frac{\sqrt{2}}{k^2}\left(n + 2 - 2\sum x\right) = \frac{\sqrt{2}}{k^2} n ,$$

$$\sum \varepsilon^2 = \frac{2}{k^4}\left(n + 2 - 4\sum x + 4\sum x^2\right) = \frac{2}{k^4} n ,$$

and

$$n \sum \varepsilon^2 = \left(\sum \varepsilon\right)^2 .$$

An interesting extension has been observed by MAULDON [8, p. 510]. If, instead of being mutually tangent, the $n+2$ spheres all intersect one another at the same

angle $\theta \leqslant \operatorname{arc\,sec}(n+1)$ or all pairs of them have the same inversive distance δ, then

$$\frac{(\sum \varepsilon)^2}{\sum \varepsilon^2} = n + 1 - \sec\theta \quad \text{or} \quad n + 1 - \operatorname{sech}\delta,$$

respectively. The above proof can be adapted by using spheres

$$\sum x^2 - 2x_\nu + 1 - r^2 = 0 \quad \text{and} \quad \sum x^2 = 1 - r^2$$

instead of

$$\sum x^2 - 2x_\nu + \tfrac{1}{2} = 0 \quad \text{and} \quad \sum x^2 = \tfrac{1}{2},$$

and then writing $\cos\theta$ or $\cosh\delta$ for $(1-r^2)/r^2$, which is Mauldon's $-\gamma$.

4. A Sequence of Successively Tangent Spheres

We have seen that, in Euclidean n-space, any $n+2$ mutually tangent $(n-1)$-spheres can be inverted into n congruent $(n-1)$-spheres sandwiched between two parallel hyperplanes. It follows that all sets of $n+2$ mutually tangent spheres are inversively equivalent, and that any set of $n+1$ mutually tangent spheres can be completed to a set of $n+2$ in just two ways. Hence, whenever $n+1$ mutually tangent spheres have been named $\sigma_1, \sigma_2, \ldots, \sigma_{n+1}$ in a definite order, they belong to just one sequence

$$\ldots, \sigma_{-1}, \sigma_0, \sigma_1, \sigma_2, \ldots$$

(extending indefinitely, backwards and forwards) such that every $n+2$ consecutive members are mutually tangent. Since every $n+2$ consecutive bends are related by a quadratic equation such as 3.1, every $n+3$ consecutive bends are related by a linear equation which can be found as follows.

Subtracting

$$n(\varepsilon_1^2 + \cdots + \varepsilon_{n+2}^2) - (\varepsilon_1 + \cdots + \varepsilon_{n+2})^2$$

from

$$n(\varepsilon_0^2 + \cdots + \varepsilon_{n+1}^2) - (\varepsilon_0 + \cdots + \varepsilon_{n+1})^2,$$

we obtain

$$(\varepsilon_0 - \varepsilon_{n+2})(n\varepsilon_0 + n\varepsilon_{n+2} - \varepsilon_0 - 2\varepsilon_1 - 2\varepsilon_2 - \cdots - 2\varepsilon_{n+1} - \varepsilon_{n+2}) = 0.$$

If $\varepsilon_0 \neq \varepsilon_{n+2}$, it follows that

$$(n-1)(\varepsilon_0 + \varepsilon_{n+2}) = 2(\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_{n+1}). \tag{4.1}$$

Considerations of continuity suffice to show that this equation still holds in the exceptional case $\varepsilon_0 = \varepsilon_{n+2}$. Since it remains valid when the subscripts are consistently increased or decreased by any integer, it provides a recursion formula enabling us to compute all the bends when $\varepsilon_1, \ldots, \varepsilon_{n+2}$ (satisfying (3.1)) are given.

5. A Spiral Sequence of Circles

When $n=2$, we have a sequence of circles such that every four consecutive members are mutually tangent, and such that every five consecutive members have bends satisfying an equation such as

$$\varepsilon_0 + \varepsilon_4 = 2(\varepsilon_1 + \varepsilon_2 + \varepsilon_3). \tag{5.1}$$

For instance, since (1.1) is satisfied by 6, 3, 2, -1, one possible sequence of bends is

$$\ldots 522, 179, 62, 23, 6, 3, 2, -1, 2, 3, 6, 23, 62, 179, 522, \ldots,$$

as in Figure 5. Again, since 1.1 is satisfied by 1, 1, 0, 0, three other possible sequences are:

$$\ldots 280, 97, 33, 12, 4, 1, 1, 0, 0, 1, 1, 4, 12, 33, 97, 280, \ldots,$$

as in Figure 6,

$$\ldots, 232, 81, 28, 9, 4, 1, 0, 1, 0, 1, 4, 9, 28, 81, 232, \ldots,$$

as in Figure 7, and

$$\ldots, 625, 216, 76, 25, 9, 4, 0, 1, 1, 0, 4, 9, 25, 76, 216, 625, \ldots,$$

as in Figure 8.

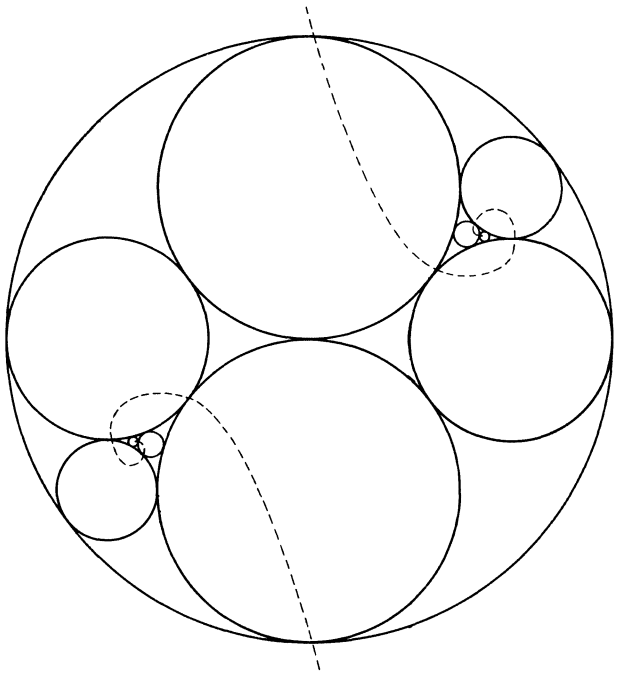

Figure 5

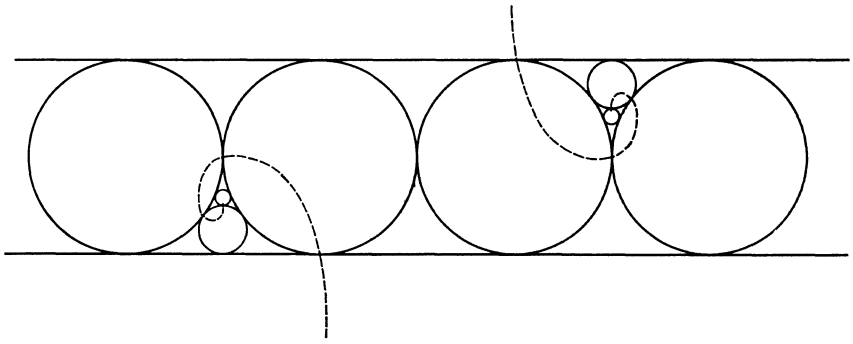

Figure 6

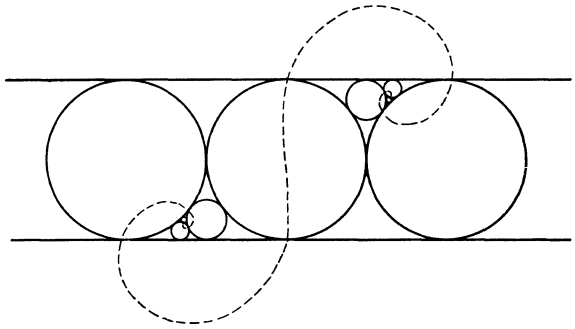

Figure 7

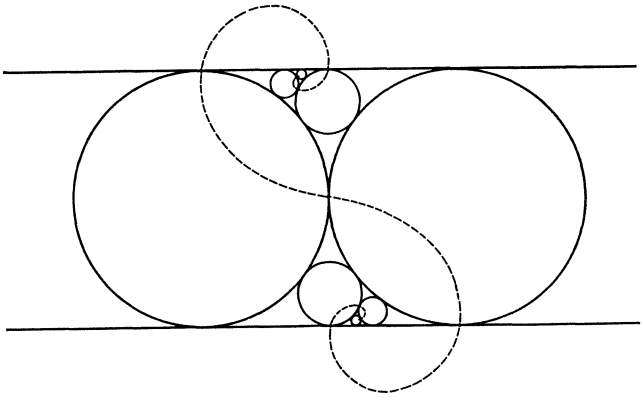

Figure 8

These special sequences illustrate an obvious corollary of (5.1):

If four consecutive bends are integers, all are integers. (5.2)

We notice, in each sequence, that $\varepsilon_{\nu+1}/\varepsilon_\nu$ seems to converge towards a value slightly less than 2.9.

The most interesting case in which the bends are not integers arises when they (and therefore the radii) belong to a geometric progression. If x is the ratio (greater than 1) of this progression, we set $\varepsilon_\nu = x^\nu$ in 5.1 to obtain

$$x^4 - 2x^3 - 2x^2 - 2x + 1 = 0$$

or

$$\{x^2 - (\sqrt{5}+1)x + 1\}\{x^2 + (\sqrt{5}-1)x + 1\} = 0. \tag{5.3}$$

In terms of the 'golden ratio' $\tau = \frac{1}{2}(\sqrt{5}+1)$, the real roots [17, p. 406] are

$$\tau + \sqrt{\tau} \approx 2.89005$$

and its reciprocal, $\tau - \sqrt{\tau}$. Thus there is a unique sequence of circles (of the kind we are considering) with their radii belonging to a G.P. with ratio $\tau + \sqrt{\tau}$ (so that each circle has about 2.89 times the radius of the preceding circle, as in Figure 9).

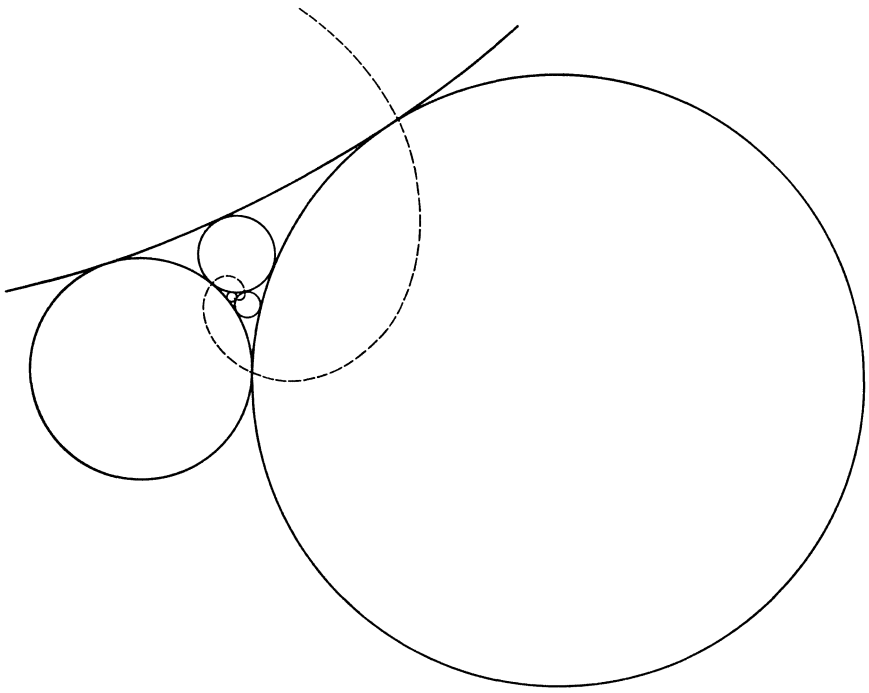

Figure 9

Since now the set of circles $\sigma_1, \sigma_2, \sigma_3 \ldots$ is directly similar to $\sigma_0, \sigma_1, \sigma_2 \ldots$, the whole sequence is transformed into itself by a dilative-rotation or spiral similarity [2, pp. 73, 125–126]: the product of a dilatation whose ratio is $x = \tau + \tau^{1/2}$ and a rotation whose angle β is the external angle at A_1 in the triangle $A_0 A_1 A_2$ formed by the centres of three consecutive circles $\sigma_0, \sigma_1, \sigma_2$. The sides $A_0 A_1$, $A_1 A_2$, $A_0 A_2$ of this triangle, being sums of pairs of radii of these circles, are proportional to $1+x$, $x+x^2$, $1+x^2$. Therefore

$$-\cos\beta = \frac{(1+x)^2 + (x+x^2)^2 - (1+x^2)^2}{2(1+x)(x+x^2)}$$
$$= \frac{x^2+1}{(x+1)^2} = \frac{2\tau x}{2\tau^2 x} = \tau^{-1},$$

$\beta \approx 128°10'$, $\sin\beta = \tau^{-1/2}$, and $e^{\beta i} = -\tau^{-1} + \tau^{-1/2} i$. It is interesting to observe that $e^{\pm\beta i}$ are the roots of the equation

$$y^2 + 2\tau^{-1} y + 1 = 0$$

which arises from the discarded factor $x^2 + 2\tau^{-1}x + 1$ in 5.3.

Using the Argand plane, we naturally identify the centre A_ν of the circle σ_ν with the complex number z^ν, where

$$z = x e^{\beta i} = (\tau + \tau^{1/2})(-\tau^{-1} + \tau^{-1/2} i) = (\tau^{1/2} + 1)(-\tau^{-1/2} + i). \tag{5.4}$$

We recall that, if x satisfies $x^2 - ax + 1 = 0$ and y satisfies $y^2 - by + 1 = 0$, then the product xy satisfies

$$z^4 - abz^3 + (a^2 + b^2 - 2)z^2 - abz + 1 = 0.$$

In the present case $a = 2\tau$ and $b = -2\tau^{-1}$; therefore z satisfies the pretty equation

$$z^4 + 4z^3 + 10z^2 + 4z + 1 = 0 \tag{5.5}$$

which can also be expressed as $(z+1)^4 + (2z)^2 = 0$.

The spiral similarity makes the centers A_ν lie on an equiangular spiral $r = e^{\theta \cot\phi}$ whose angle ϕ is given by $x = e^{\beta \cot\phi}$, that is,

$$\left.\begin{aligned} \tan\phi &= \frac{\beta}{\log x} = \frac{\arccos(-\tau^{-1})}{\log(\tau + \tau^{1/2})} \\ &\approx \frac{3.1416 - 0.9045}{1.0613} \approx 2.108, \qquad \phi \approx 64°37'. \end{aligned}\right\} \tag{5.6}$$

Like any other sequence of circles in which every four consecutive members are mutually tangent, Figure 9 can be transformed into Figure 6 (or into its reflected image) by inversion in a circle whose centre is the point of contact of two consecutive

circles. Therefore all such sequences of circles are inversively equivalent. Although centres do not invert into centres, points of contact do invert into points of contact. Thus the inversively significant equiangular spiral is not the one passing through the centres but the similar (and therefore congruent) one passing through the successive points of contact. Such points of contact always lie on a *loxodrome* [2a, pp. 144, 450; 3, pp. 9–10] which, being the inverse of an equiangular spiral, has two poles (points of accumulation of the circles), one arising from the pole of the spiral and the other from the point at infinity. The inversion transforms the spiral similarity into a *loxodromic homography* [2, pp. 91, 146], which makes it natural to call this arrangement of circles a *loxodromic sequence*.

Conversely, the *spiral sequence*, with radii in a geometric progression, arises from the loxodromic sequence by inversion in a circle whose centre is at one of the two points of accumulation.

Of course, there are also points of contact of non-consecutive circles, such as σ_0 and σ_2, or σ_0 and σ_3. Inversion in a circle with such a centre yields Figure 7 or Figure 8. Although these two figures are different loxodromic sequences of circles, their loxodromes are similar. This happens because, among the various loxodromes of a given 'pitch' (arising from an equiangular spiral with a given angle ϕ by different circles of inversion), only two are centrally symmetric: the one that passes through its centre of symmetry, as in Figures 7 and 8, and the one that has an asymptote, as in Figures 5 and 6. The latter kind of loxodrome has an amusing property: every point on it is the centre of a circle that inverts it into an oppositely congruent curve!

Notice also that the outermost circle in Figure 5 inverts this loxodromic sequence of circles into Figure 7. Similarly, Figures 6 and 8 can be interchanged by inversion in a centrally located circle.

6. Inversive Distances

The sequence of circles, every four consecutive members mutually tangent, may be regarded as belonging to *inversive* geometry. In this sense it is unique: the distinction between Figures 5, 6, 7, 8, 9 disappears. Although the inversive distances among the consecutive circles σ_0, σ_1, σ_2, σ_3 are all zero [4, p. 124], it is pertinent to ask what is the inversive distance between σ_0 and σ_4 (or any other two that are as close as possible without being tangent). This is easily computed from Figure 6 or 7 or 8, where we can take one of the two circles to have bend 4 (that is, radius $\frac{1}{4}$) while the other is a straight line at distance $2-\frac{1}{4}$ from the centre. It follows [4, pp. 131, 176 (ex. 4)] that the inversive distance δ is given by

$$\cosh \delta = 7,$$

whence $\cosh \frac{1}{2}\delta = 2$, $\delta = 2\log(2+\sqrt{3})$. Thus, if δ_ν is the inversive distance between

circles σ_μ and $\sigma_{\mu+\nu}$, we have

$$\delta_1 = \delta_2 = \delta_3 = 0, \qquad \cosh \tfrac{1}{2}\delta_4 = 2$$

and similarly (from Figure 8, again)

$$\cosh \tfrac{1}{2}\delta_5 = 3, \qquad \cosh \tfrac{1}{2}\delta_6 = 5.$$

It is natural to conjecture that $\cosh \delta_\nu$ will always be an integer: a nice inversive counterpart for (5.2).

This conjecture can, in fact, be proved by using the spiral sequence (Figure 9). For such circles, the centres A_ν are represented by the complex numbers z^ν (see (5.4)). The radii $r_\nu = \varepsilon_\nu^{-1}$ are given by

$$r_\nu = x^\nu r_0, \qquad (x+1) r_0 = r_1 + r_0 = A_0 A_1 = |z-1|.$$

Since $x^2 - 2\tau x + 1 = 0$, we have

$$r_0^2 = \frac{|z-1|^2}{(x+1)^2} = \frac{x^2 + 1 - 2x\cos\beta}{2x + 2\tau x} = \frac{2\tau x + 2\tau^{-1} x}{2\tau^2 x} = \frac{\sqrt{5}}{\tau^2},$$

$$r_0 = 5^{1/4}\tau^{-1}, \qquad r_\nu = 5^{1/4}\tau^{-1} x^\nu.$$

Since

$$A_0 A_\nu^2 = |z^\nu - 1|^2 = x^{2\nu} + 1 - 2x^\nu \cos\nu\beta,$$

it follows [4, p. 129] that

$$\begin{aligned}
\cosh \delta_\nu &= \frac{A_0 A_\nu^2 - r_0^2 - r_\nu^2}{2 r_0 r_\nu} = \frac{x^{2\nu} + 1 - 2x^\nu \cos\nu\beta}{2\sqrt{5}\tau^{-2} x^\nu} - \frac{1 + x^{2\nu}}{2x^\nu} \\
&= \frac{(\tau^2 - \sqrt{5})(x^\nu + x^{-\nu})}{2\sqrt{5}} - \frac{\tau^2 \cos\nu\beta}{\sqrt{5}} \\
&= \frac{\tau^{-2}}{2\sqrt{5}} \{(\tau + \tau^{1/2})^\nu + (\tau - \tau^{1/2})^\nu\} - \frac{\tau^2}{2\sqrt{5}} \{(-\tau^{-1} + \tau^{-1/2} i)^\nu \\
&\qquad\qquad + (-\tau^{-1} - \tau^{-1/2} i)^\nu\} \\
&= \frac{\tau^{-2}}{\sqrt{5}} \left\{\tau^\nu + \binom{\nu}{2}\tau^{\nu-1} + \binom{\nu}{4}\tau^{\nu-2} + \cdots\right\} \\
&\qquad - \frac{\tau^2}{\sqrt{5}} \left\{(-\tau^{-1})^\nu + \binom{\nu}{2}(-\tau^{-1})^{\nu-1} + \binom{\nu}{4}(-\tau^{-1})^{\nu-2} + \cdots\right\} \\
&= \frac{\tau^{\nu-2} - (-\tau)^{-\nu+2}}{\sqrt{5}} + \binom{\nu}{2}\frac{\tau^{\nu-3} - (-\tau)^{-\nu+3}}{\sqrt{5}} \\
&\qquad\qquad + \binom{\nu}{4}\frac{\tau^{\nu-4} - (-\tau)^{-\nu+4}}{\sqrt{5}} + \cdots \\
&= f_{\nu-2} + \binom{\nu}{2} f_{\nu-3} + \binom{\nu}{4} f_{\nu-4} + \cdots,
\end{aligned}$$

where f_κ is the κth Fibonacci number [2, p. 167, 11.49]. This proves that $\cosh \delta_\nu$ is indeed an integer. In particular,

$$\begin{aligned}
\cosh \delta_5 &= f_3 + 10 f_2 + 5 f_1 = 2 + 10 + 5 = 17\,,\\
\cosh \delta_6 &= f_4 + 15 f_3 + 15 f_2 + f_1 = 3 + 30 + 15 + 1 = 49\,,\\
\cosh \delta_7 &= f_5 + 21 f_4 + 35 f_3 + 7 f_2 = 5 + 63 + 70 + 7 = 145\,,\\
\cosh \delta_8 &= f_6 + 28 f_5 + 70 f_4 + 28 f_3 + f_2 = 8 + 140 + 210 + 56 + 1 = 415\,.
\end{aligned}$$

As ν increases, $\exp(\delta_{\nu+1} - \delta_\nu)$, like $\cosh \delta_{\nu+1}/\cosh \delta_\nu$, tends to x. Therefore $\delta_{\nu+1} - \delta_\nu$ tends to $\log x \approx 1.06126$ and δ_ν is of the same order of magnitude as ν itself. The first eight cases are

$$\delta_1 = \delta_2 = \delta_3 = 0\,, \quad \delta_4 \approx 2.634\,, \quad \delta_5 \approx 3.53\,, \quad \delta_6 \approx 4.58\,, \quad \delta_7 \approx 5.67\,, \quad \delta_8 \approx 6.72\,.$$

The last of these is already in good agreement with the asymptotic expression

$$\begin{aligned}
\delta_\nu \sim \log \frac{\tau^{-2} x^\nu}{\sqrt{5}} &= \nu \log x - 2 \log \tau - \tfrac{1}{2} \log 5\\
&\approx 1.06126\, \nu - 1.7671\,.
\end{aligned}$$

(Note: $\delta_{28} \approx 27.95$, $\delta_{29} \approx 29.01$.)

7. A Spiral Sequence of Spheres

When $n = 3$, § 4 yields a sequence of ordinary spheres such that every five consecutive members are mutually tangent and such that every six consecutive members have bends satisfying an equation such as

$$\varepsilon_0 + \varepsilon_5 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4\,. \tag{7.1}$$

For instance, since Soddy's equation

$$3(\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \varepsilon_4^2 + \varepsilon_5^2) = (\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 + \varepsilon_5)^2$$

is satisfied by 3, 3, 2, 2, -1, one possible sequence of bends is

$$\ldots, 155, 81, 42, 23, 14, 5, 3, 3, 2, -1, 2, 3, 3, 5, 14, 23, 42, 81, 155, \ldots,$$

as in his *bowl of integers* [14]. Since the same equation is satisfied by 1, 1, 1, 0, 0, another possible sequence is

$$\ldots, 244, 129, 69, 37, 19, 10, 6, 3, 1, 1, 1, 0, 0, 1, 1, 1, 3, 6, 10, 19, 37, 69, 129, 244, \ldots,$$

as when six congruent spheres are sandwiched between two parallel planes.

Clearly, 5.2 remains valid (with 'five' instead of 'four'). But now, in each sequence, $\varepsilon_{\nu+1}/\varepsilon_\nu$ converges towards a value slightly less than 1.9.

As before, it is possible for the bends (and therefore the radii) to form a geometric progression. Setting $\varepsilon_\nu = x^\nu$ in 7.1, we obtain

$$x^5 - x^4 - x^3 - x^2 - x + 1 = 0,$$

that is,

$$(x+1)\{x^2 - (\sqrt{2}+1)x + 1\}\{x^2 + (\sqrt{2}-1)x + 1\} = 0.$$

The only non-trivial real roots are

$$\tfrac{1}{2}(\sqrt{2} + 1 + \sqrt{2\sqrt{2} - 1}) = 1.88\ldots$$

and its reciprocal.

As in the two-dimensional case, the whole sequence is transformed into itself by a dilative-rotation or *spiral similarity* [2, pp. 102, 326]. Therefore the points of contact of consecutive spheres lie on a concho-spiral, which is a curve drawn on a right circular cone so as to cut the generators at a constant angle. It follows that any other sequence of spheres, with every five consecutive members mutually tangent, has just two points of accumulation, and that the points of contact of consecutive spheres lie on a 'twisted loxodrome' which, being the inverse of a concho-spiral, lies on a bi-nodal cyclide [12, p. 135 (§ 567)] cutting the lines of curvature (of one system) at this same constant angle.

8. The Extension to n Dimensions

Many of the above results (though not 5.2) remain valid when $n \geqslant 3$. For instance, the four-dimensional case of (4.1) is

$$\varepsilon_0 + \varepsilon_6 = \tfrac{2}{3}(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 + \varepsilon_5),$$

yielding a sequence of bends

$$\ldots, \tfrac{8}{3}, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, \tfrac{8}{3}, \ldots,$$

Setting $\varepsilon_\nu = x^\nu$ in (4.1), we see that the ratio of the geometric progression (of bends or radii) is given by

$$n(x^{n+2} + 1) = x^{n+2} + 2(x^{n+1} + \cdots + x) + 1,$$

whence

$$\begin{aligned} n(x-1)(x^{n+2}+1) &= (x-1)(x^{n+2}+1) + 2x(x^{n+1}-1) \\ &= (x+1)(x^{n+2}-1). \end{aligned}$$

Thus

$$\frac{x^{n+2}-1}{x^{n+2}+1} = n\,\frac{x-1}{x+1}.$$

If there is a root $x > 1$, it must be expressible as $x = e^{2t}$, where $t > 0$. This substitution

yields

$$\tanh(n+2)\,t = n \tanh t\,. \tag{8.1}$$

When t increases steadily through positive values (while n remains fixed),

$$\frac{\tanh(n+2)\,t}{\tanh t}$$

decreases steadily from $n+2$ to 1, and must therefore attain the value n just once. Hence there exists, for every n, a spiral sequence of spheres, every $n+2$ consecutive members mutually tangent, with their radii in a geometric progression.

Let t_n denote the unique positive root of the equation 8.1. We have seen that $t_3 < t_2$. Since

$$n \tanh t_n = \tanh(n+2)\,t_n < 1\,,$$

t_n tends to zero as n tends to infinity. Can we be sure that t_n tends *steadily* to zero? The expression for dt/dn is hideously complicated, but J. B. WILKER has kindly supplied the following simple proof that $t_n < t_{n-1}$.

If s and t are positive,

$$\tanh(s+t) = \frac{\tanh s + \tanh t}{1 + \tanh s \tanh t} < \tanh s + \tanh t\,.$$

Therefore

$$\frac{\tanh(n+3)\,t_n}{\tanh t_n} < \frac{\tanh(n+2)\,t_n}{\tanh t_n} + 1 = n+1\,.$$

Writing $n-1$ for n, we deduce

$$\frac{\tanh(n+2)\,t_{n-1}}{\tanh t_{n-1}} < n\,.$$

In the steady variation of $\tanh(n+2)\,t/\tanh t$ from $n+2$ down to 1, the value n occurs before a value less than n; therefore

$$t_n < t_{n-1}\,.$$

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